

# LEXAN™ COPOLYMER EXL5689

## DESCRIPTION

Lexan EXL5689 polycarbonate (PC) resin is a GF reinforced, UV stabilized, flame retardant injection molding copolymer blend. This medium flow resin features UL94 V0 @ 1.5mm flame retardancy based on non-chlorine, non-bromine FR agents with excellent processability and improved release performance. Lexan EXL5689 resin offers much improved impact strength and ductility over conventional GF reinforced PC resins. This product is an excellent candidate for a broad range of applications, including electrical and electronic enclosures among others.

| GENERAL INFORMATION   |                                                        |
|-----------------------|--------------------------------------------------------|
| Features              | Flame Retardant, Chemical Resistance, Impact resistant |
| Fillers               | Glass Fiber                                            |
| Polymer Types         | Polycarbonate (PC)                                     |
| Processing Techniques | Injection Molding                                      |

  

| INDUSTRY                   | SUB INDUSTRY                                |
|----------------------------|---------------------------------------------|
| Building and Construction  | Building Component, Water Management        |
| Electrical and Electronics | Mobile Phone - Computer - Tablets, Lighting |
| Industrial                 | Electrical, Defense                         |
| Mass Transportation        | Rail                                        |

## TYPICAL PROPERTY VALUES

Revision 20220720

| PROPERTIES                                   | TYPICAL VALUES | UNITS             | TEST METHODS |
|----------------------------------------------|----------------|-------------------|--------------|
| <b>MECHANICAL <sup>(1)</sup></b>             |                |                   |              |
| Tensile Stress, yld, Type I, 5 mm/min        | 55             | MPa               | ASTM D638    |
| Tensile Stress, brk, Type I, 5 mm/min        | 44             | MPa               | ASTM D638    |
| Tensile Strain, yld, Type I, 5 mm/min        | 4.4            | %                 | ASTM D638    |
| Tensile Strain, brk, Type I, 5 mm/min        | 15             | %                 | ASTM D638    |
| Tensile Modulus, 5 mm/min                    | 3500           | MPa               | ASTM D638    |
| Flexural Stress, yld, 1.3 mm/min, 50 mm span | 97             | MPa               | ASTM D790    |
| Flexural Modulus, 1.3 mm/min, 50 mm span     | 3150           | MPa               | ASTM D790    |
| Tensile Stress, yield, 5 mm/min              | 54             | MPa               | ISO 527      |
| Tensile Stress, break, 5 mm/min              | 46             | MPa               | ISO 527      |
| Tensile Strain, yield, 5 mm/min              | 4.4            | %                 | ISO 527      |
| Tensile Strain, break, 5 mm/min              | 13             | %                 | ISO 527      |
| Tensile Modulus, 1 mm/min                    | 3600           | MPa               | ISO 527      |
| Flexural Stress, yield, 2 mm/min             | 96             | MPa               | ISO 178      |
| Flexural Modulus, 2 mm/min                   | 3400           | MPa               | ISO 178      |
| <b>IMPACT <sup>(1)</sup></b>                 |                |                   |              |
| Izod Impact, notched, 23°C                   | 340            | J/m               | ASTM D256    |
| Izod Impact, notched, -30°C                  | 150            | J/m               | ASTM D256    |
| Instrumented Dart Impact Total Energy, 23°C  | 40             | J                 | ASTM D3763   |
| Izod Impact, unnotched 80*10*3 +23°C         | NB             | kJ/m <sup>2</sup> | ISO 180/1U   |
| Izod Impact, unnotched 80*10*3 -30°C         | NB             | kJ/m <sup>2</sup> | ISO 180/1U   |

| PROPERTIES                                          | TYPICAL VALUES                    | UNITS                   | TEST METHODS   |
|-----------------------------------------------------|-----------------------------------|-------------------------|----------------|
| Izod Impact, notched 80*10*3 +23°C                  | 25                                | kJ/m <sup>2</sup>       | ISO 180/1A     |
| Izod Impact, notched 80*10*3 -30°C                  | 10                                | kJ/m <sup>2</sup>       | ISO 180/1A     |
| Charpy 23°C, V-notch Edgew 80*10*3 sp=62mm          | 30                                | kJ/m <sup>2</sup>       | ISO 179/1eA    |
| Charpy -30°C, V-notch Edgew 80*10*3 sp=62mm         | 15                                | kJ/m <sup>2</sup>       | ISO 179/1eA    |
| Charpy 23°C, Unnotch Edgew 80*10*3 sp=62mm          | NB                                | kJ/m <sup>2</sup>       | ISO 179/1eU    |
| Charpy -30°C, Unnotch Edgew 80*10*3 sp=62mm         | NB                                | kJ/m <sup>2</sup>       | ISO 179/1eU    |
| THERMAL <sup>(1)</sup>                              |                                   |                         |                |
| Vicat Softening Temp, Rate B/50                     | 146                               | °C                      | ASTM D1525     |
| CTE, -40°C to 40°C, flow                            | 4.7E-05                           | 1/°C                    | ASTM E831      |
| CTE, -40°C to 40°C, xflow                           | 7.E-05                            | 1/°C                    | ASTM E831      |
| CTE, -40°C to 40°C, flow                            | 4.7E-05                           | 1/°C                    | ISO 11359-2    |
| CTE, -40°C to 40°C, xflow                           | 7.E-05                            | 1/°C                    | ISO 11359-2    |
| Ball Pressure Test, 125°C +/- 2°C                   | passes                            | -                       | IEC 60695-10-2 |
| Vicat Softening Temp, Rate B/50                     | 145                               | °C                      | ISO 306        |
| Vicat Softening Temp, Rate B/120                    | 146                               | °C                      | ISO 306        |
| HDT/Ae, 1.8 MPa Edgew 120*10*4 sp=100mm             | 132                               | °C                      | ISO 75/Ae      |
| Relative Temp Index, Elec <sup>(2)</sup>            | 130                               | °C                      | UL 746B        |
| Relative Temp Index, Mech w/impact <sup>(2)</sup>   | 120                               | °C                      | UL 746B        |
| Relative Temp Index, Mech w/o impact <sup>(2)</sup> | 130                               | °C                      | UL 746B        |
| PHYSICAL <sup>(1)</sup>                             |                                   |                         |                |
| Specific Gravity                                    | 1.26                              | -                       | ASTM D792      |
| Mold Shrinkage, flow, 3.2 mm <sup>(3)</sup>         | 0.2 – 0.6                         | %                       | SABIC method   |
| Melt Flow Rate, 300°C/ 1.2 kgf                      | 9                                 | g/10 min                | ASTM D1238     |
| Density                                             | 1.26                              | g/cm <sup>3</sup>       | ISO 1183       |
| Water Absorption, (23°C/saturated)                  | 0.35                              | %                       | ISO 62-1       |
| Moisture Absorption (23°C / 50% RH)                 | 0.15                              | %                       | ISO 62         |
| Melt Volume Rate, MVR at 300°C/ 1.2 kg              | 8                                 | cm <sup>3</sup> /10 min | ISO 1133       |
| ELECTRICAL <sup>(2)</sup>                           |                                   |                         |                |
| High Voltage Arc Track Rate {PLC}                   | 4                                 | PLC Code                | UL 746A        |
| Comparative Tracking Index (UL) {PLC}               | 3                                 | PLC Code                | UL 746A        |
| Volume Resistivity                                  | >1.E+15                           | Ω.cm                    | IEC 60093      |
| Dielectric Strength in oil, 1.5mm                   | 27                                | kV/mm                   | IEC 60243-1    |
| Comparative Tracking Index                          | 175                               | V                       | IEC 60112      |
| Hot-Wire Ignition (HWI), PLC 2                      | ≥3                                | mm                      | UL 746A        |
| Hot-Wire Ignition (HWI), PLC 3                      | ≥0.75                             | mm                      | UL 746A        |
| High Amp Arc Ignition (HAI), PLC 1                  | ≥1                                | mm                      | UL 746A        |
| High Amp Arc Ignition (HAI), PLC 4                  | ≥0.75                             | mm                      | UL 746A        |
| Dielectric Constant, 1.1 GHz <sup>(1)</sup>         | 2.95                              | -                       | SABIC method   |
| Dissipation Factor, 1.1 GHz <sup>(1)</sup>          | 0.0058                            | -                       | SABIC method   |
| Dielectric Constant, 1.9 GHz <sup>(1)</sup>         | 3                                 | -                       | SABIC method   |
| Dissipation Factor, 1.9 GHz <sup>(1)</sup>          | 0.0062                            | -                       | SABIC method   |
| FLAME CHARACTERISTICS <sup>(2)</sup>                |                                   |                         |                |
| UL Yellow Card Link <sup>(2)</sup>                  | <a href="#">E45329-561961</a>     | -                       | -              |
| UL Yellow Card Link 2                               | <a href="#">E121562-103854053</a> | -                       | -              |
| UL Yellow Card Link 3                               | <a href="#">E207780-101033175</a> | -                       | -              |

| PROPERTIES                                  | TYPICAL VALUES | UNITS | TEST METHODS   |
|---------------------------------------------|----------------|-------|----------------|
| UL Recognized, 94HB Flame Class Rating      | ≥0.75          | mm    | UL 94          |
| UL Recognized, 94V-0 Flame Class Rating     | ≥1.5           | mm    | UL 94          |
| UL Recognized, 94V-1 Flame Class Rating     | ≥1.2           | mm    | UL 94          |
| Glow Wire Ignitability Temperature, 3.0 mm  | 825            | °C    | IEC 60695-2-13 |
| Glow Wire Ignitability Temperature, 1.5 mm  | 825            | °C    | IEC 60695-2-13 |
| Glow Wire Ignitability Temperature, 1.2 mm  | 825            | °C    | IEC 60695-2-13 |
| Glow Wire Ignitability Temperature, 1.1 mm  | 825            | °C    | IEC 60695-2-13 |
| Glow Wire Ignitability Temperature, 1.0 mm  | 825            | °C    | IEC 60695-2-13 |
| Glow Wire Ignitability Temperature, 0.75 mm | 825            | °C    | IEC 60695-2-13 |
| Glow Wire Flammability Index, 3.0 mm        | 960            | °C    | IEC 60695-2-12 |
| Glow Wire Flammability Index, 1.5 mm        | 960            | °C    | IEC 60695-2-12 |
| Glow Wire Flammability Index, 1.2 mm        | 960            | °C    | IEC 60695-2-12 |
| Glow Wire Flammability Index, 1.1 mm        | 960            | °C    | IEC 60695-2-12 |
| Glow Wire Flammability Index, 1.0 mm        | 825            | °C    | IEC 60695-2-12 |
| Glow Wire Flammability Index, 0.75 mm       | 825            | °C    | IEC 60695-2-12 |
| UV-light, water exposure/immersion          | F1             | -     | UL 746C        |
| <b>INJECTION MOLDING <sup>(4)</sup></b>     |                |       |                |
| Drying Temperature                          | 120            | °C    |                |
| Drying Time                                 | 3 – 4          | Hrs   |                |
| Drying Time (Cumulative)                    | 48             | Hrs   |                |
| Maximum Moisture Content                    | 0.02           | %     |                |
| Melt Temperature                            | 270 – 330      | °C    |                |
| Nozzle Temperature                          | 265 – 325      | °C    |                |
| Front - Zone 3 Temperature                  | 270 – 330      | °C    |                |
| Middle - Zone 2 Temperature                 | 260 – 320      | °C    |                |
| Rear - Zone 1 Temperature                   | 250 – 310      | °C    |                |
| Mold Temperature                            | 80 – 115       | °C    |                |
| Back Pressure                               | 0.3 – 0.7      | MPa   |                |
| Screw Speed                                 | 40 – 70        | rpm   |                |
| Shot to Cylinder Size                       | 40 – 60        | %     |                |
| Vent Depth                                  | 0.025 – 0.076  | mm    |                |

- (1) The information stated on Technical Datasheets should be used as indicative only for material selection purposes and not be utilized as specification or used for part or tool design.
- (2) UL Ratings shown on the technical datasheet might not cover the full range of thicknesses and colors. For details, please see the UL Yellow Card.
- (3) Measurements made from laboratory test coupon. Actual shrinkage may vary outside of range due to differences in processing conditions, equipment, part geometry and tool design. It is recommended that mold shrinkage studies be performed with surrogate or legacy tooling prior to cutting tools for new molded article.
- (4) Injection Molding parameters are only mentioned as general guidelines. These may not apply or may need adjustment in specific situations such as low shot sizes, large part molding, thin wall molding and gas-assist molding.

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